

NCA Sponsored Training

Day 3 – Morning Session (Lab)

Network Security 1

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Outline



1. Packet Sniffing

- *Definition & Purpose*

2. Wireshark

- *Introduction*
- *GUI & Interface*

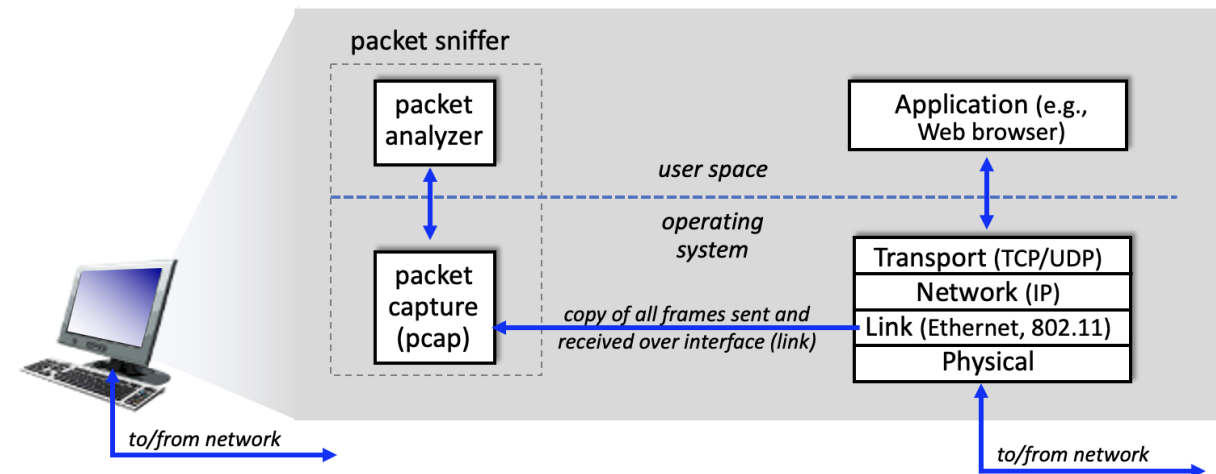
Slides source: *Computer Networks, A Top-down Approach, 8th ed., J.F. Kurose and K.W. Ross, Addison-Wesley/Pearson, 2020.*



Packet Sniffing



- The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**.
 - captures copy of (“sniffs”) messages being sent/received from/by your computer
 - store and/or display the contents of the various protocol fields
 - Passive entity i.e., observer, not transmitter of packets
 - consists of two parts:
 - **Packet capture library**
 - captures all Wifi/Ethernet frames
 - **Packet analyzer**
 - displays protocols field msgs





- packet analyzer that uses a packet capture (pcap) library in your computer
- *free and open source*
- deep inspection of hundreds of protocols
- live capture and offline analysis
- multi-platform: Runs on Windows, Linux, OS X, FreeBSD, NetBSD, and many others
- getting Wireshark: <http://www.wireshark.org/download.html>
- detailed FAQs: <http://www.wireshark.org/docs/man-pages/>



Wireshark – Welcom Screen



Apply a display filter ... <?>

Welcome to Wireshark

Capture

...using this filter: All interfaces shown

Wi-Fi: en0

awdl0

Loopback: lo0

ap1

llw0

utun0

utun1

utun2

utun3

anpi2

anpi1

anpi0

Ethernet Adapter (en5): en5

Ethernet Adapter (en6): en6

Ethernet Adapter (en7): en7

Thunderbolt 1: en1

Thunderbolt 2: en2

Thunderbolt 3: en3

Thunderbolt Bridge: bridge0

gif0

stf0

Cisco remote capture: ciscodump

Random packet generator: randpkt

SSH remote capture: sshdump

UDP Listener remote capture: udpdump

Wi-Fi remote capture: wifidump

Learn

User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate

You are running Wireshark 4.2.0 (v4.2.0-0-g54eedfc63953). You receive automatic updates.

Ready to load or capture

No Packets

Profile: Default

List of Interfaces



Wireshark – Getting Started



command menus

display filter specification

listing of captured packets

Details of selected packet

packet content (in hexadecimal and ASCII)

Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Wi-Fi: en0

Apply a display filter ... <%%/>

No.	Time	Source	Destination	Protocol	Length	Info
55	9.523398	128.119.245.12	192.168.0.15	TCP	74	80 → 55621 [SYN, ACK, ECN] Seq=0 Ack=1 Win=28
56	9.523490	192.168.0.15	128.119.245.12	TCP	66	55621 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0
57	9.523911	192.168.0.15	128.119.245.12	HTTP	697	GET /wireshark-labs/INTRO-wireshark-file1.htm
58	9.525361	128.119.245.12	192.168.0.15	TCP	74	80 → 55622 [SYN, ACK, ECN] Seq=0 Ack=1 Win=28
59	9.525418	192.168.0.15	128.119.245.12	TCP	66	55622 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0
60	9.610608	35.166.241.18	192.168.0.15	TLSv1.2	117	Change Cipher Spec, Encrypted Handshake Message
61	9.610613	128.119.245.12	192.168.0.15	TCP	66	80 → 55621 [ACK] Seq=1 Ack=632 Win=30336 Len=0
62	9.610732	192.168.0.15	35.166.241.18	TCP	66	55620 → 443 [ACK] Seq=644 Ack=3451 Win=131008
63	9.616465	128.119.245.12	192.168.0.15	HTTP	305	HTTP/1.1 304 Not Modified
64	9.616562	192.168.0.15	128.119.245.12	TCP	66	55621 → 80 [ACK] Seq=632 Ack=240 Win=131520 Len=0
65	9.690975	192.168.0.15	192.168.0.254	DNS	99	Standard query 0xe646 AAAA captive-cdn.origin-apple.co
66	9.691048	192.168.0.15	192.168.0.254	DNS	99	Standard query 0x8a27 A captive-cdn.origin-apple.co

▶ Frame 57: 697 bytes on wire (5576 bits), 697 bytes captured (5576 bits) on interface en0, id 0

▶ Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: FreeboxS_aa:0f:e4 (00:24:d4:aa:0f:e4)

▶ Internet Protocol Version 4, Src: 192.168.0.15, Dst: 128.119.245.12

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

▶ Differentiated Services Field: 0x02 (DSCP: CS0, ECN: ECT(0))

Total Length: 683

Identification: 0x0000 (0)

▶ Flags: 0x4000, Don't fragment

Fragment offset: 0

Time to live: 64

0010	02 ab 00 00 40 00 40 06	02 10 c0 a8 00 0f 80 77	...@... ..w
0020	f5 0c d9 45 00 50 15 e5	a0 3d df b1 a1 4c 80 18	...E.P... ..L..
0030	08 0a 20 1d 00 00 01 01	08 0a 0a 95 91 2a d9 eb	*..
0040	ac e2 47 45 54 20 2f 77	69 72 65 73 68 61 72 6b	..GET /w ireshark
0050	2d 6c 61 62 73 2f 49 4e	54 52 4f 2d 77 69 72 65	-labs/IN TRO-wire
0060	73 68 61 72 6b 2d 66 69	6c 65 31 2e 68 74 6d 6c	shark-fi le1.html
0070	20 48 54 54 50 2f 31 2e	31 0d 0a 48 6f 73 74 3a	HTTP/1. 1..Host:
0080	20 67 61 69 61 2e 63 73	2e 75 6d 61 73 73 2e 65	gaia.cs .umass.e
0090	64 75 0d 0a 55 73 65 72	2d 41 67 65 6e 74 3a 20	du..User -Agent:
00a0	4d 6f 7a 69 6c 6c 61 2f	35 2e 30 20 28 4d 61 63	Mozilla/ 5.0 (Mac
00b0	69 6e 74 6f 73 68 3b 20	49 6e 74 65 6c 20 4d 61	intosh; Intel Ma
00c0	63 20 4f 53 20 58 20 31	30 2e 31 35 3b 20 72 76	c OS X 1 0.15; rv
00d0	3a 37 35 2e 30 29 20 47	65 63 6b 6f 2f 32 30 31	:75.0) Gecko/201
00e0	30 30 31 30 31 20 46 69	72 65 66 6f 78 2f 37 35	00101 Fi refox/75
00f0	2e 30 0d 0a 41 63 63 65	70 74 3a 20 74 65 78 74	.0..Acce pt: text

Time to live (ip.ttl), 1 byte

Packets: 90 · Displayed: 90 (100.0%) · Dropped: 0 (0.0%) · Profile: Default

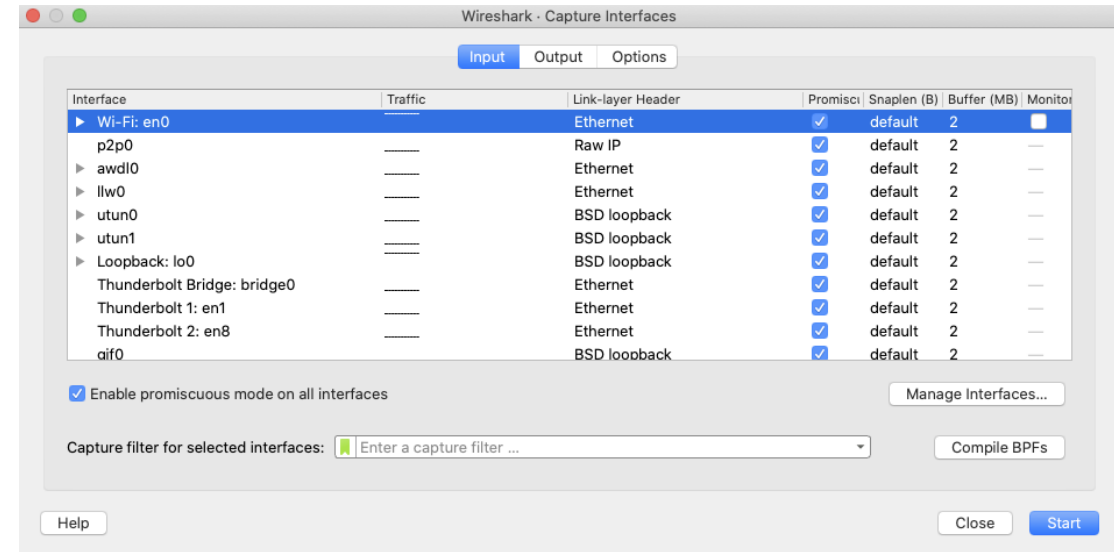


Wireshark – Test Run



- Start up your favorite web browser, which will display your selected homepage
- Start up the Wireshark software.
- Select interface from "Capture Options"
 - Wifi or Ethernet
- Select "Start capturing packets"
- While Wireshark is running, enter the URL:

<http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>

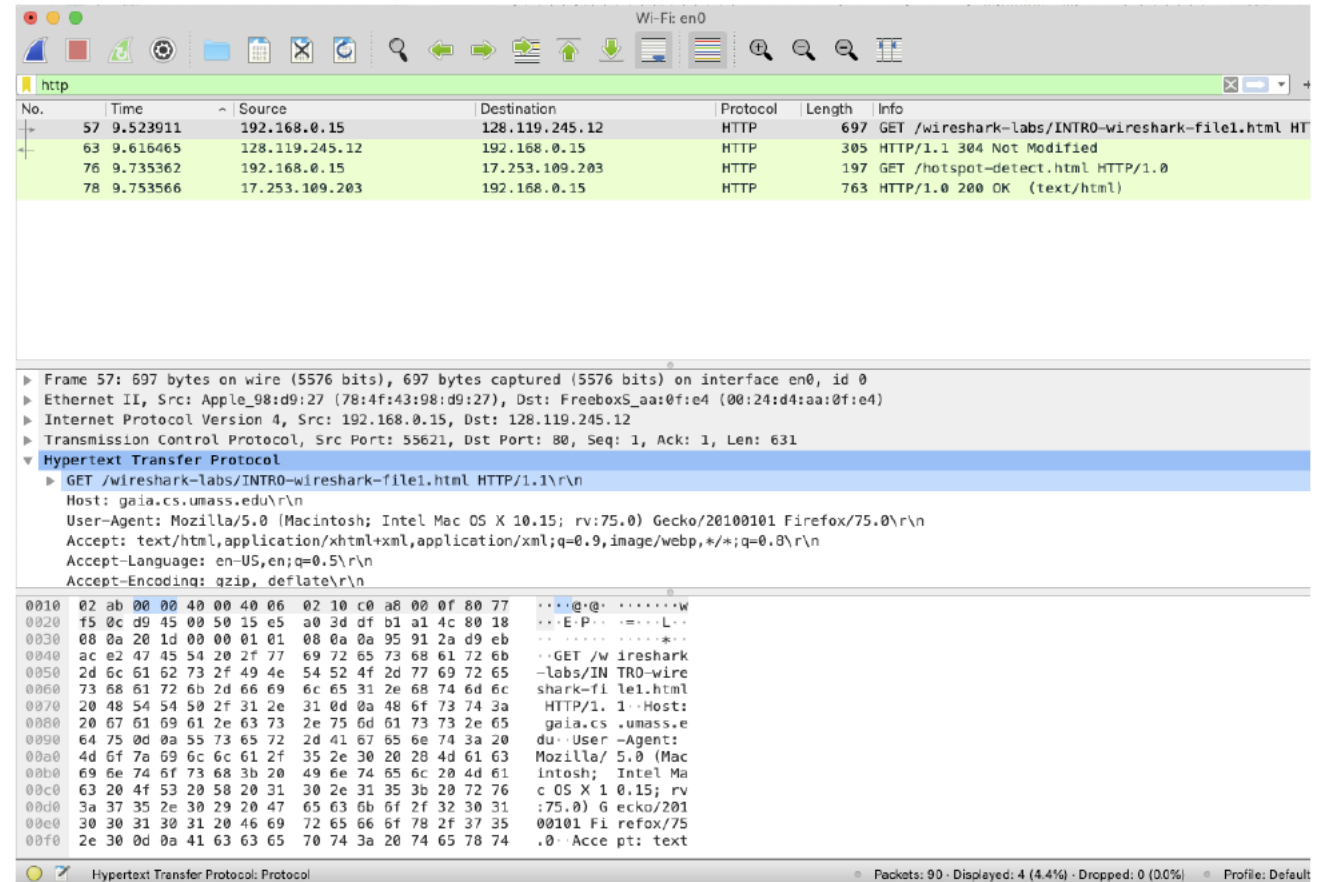




Wireshark – Test Run



- After your browser has displayed the **INTRO-wireshark-file1.html** page
Select “Stop capturing packets”
- Type in “http”
(lower cases without quotes)
in “display filter specification”
- Observe the “listing” and “details”
of captured packets
- Find the HTTP GET message that
was sent from your computer
to the gaia.cs.umass.edu HTTP server

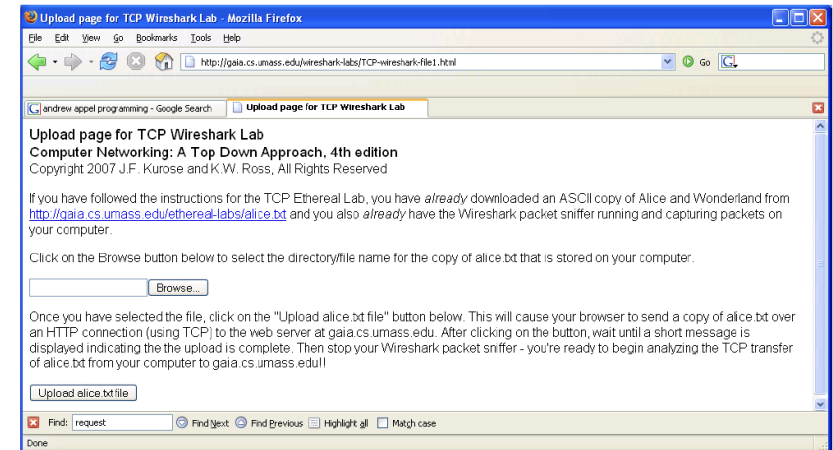




Wireshark - TCP Exercise



- Start up your web browser. Go the <http://gaia.cs.umass.edu/wireshark-labs/alice.txt> and retrieve an ASCII copy of Alice in Wonderland. Store this as a .txt file somewhere on your computer.
- Next go to <http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html>.
- You should see a screen that looks like Figure →
- Use the Browse button in this form to enter the name of the file on your computer containing Alice in Wonderland (or manually). Don't yet press the **“Upload alice.txt file”** button.
- Startup Wireshark and begin packet capture
- Returning to your browser, press the “Upload alice.txt file” button to upload the file to the gaia.cs.umass.edu server. Once the file has been uploaded, a short congratulations message will be displayed in your browser window.
- Stop Wireshark packet capture.





Wireshark - TCP Exercise (contd..)



- Answer the following questions, by opening the Wireshark captured packet file *tcpethereal-trace-1* in <http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip>
 - What is the IP address and TCP port number used by the client computer
 - What is the IP and Port address of gaia.cs.umass.edu?
 - What is the sequence number of the **TCP SYN** segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu?
 - What is it in the segment that identifies the segment as a **SYN** segment?
 - What is the sequence number of the **SYNACK** segment sent by gaia.cs.umass.edu to the client computer in reply to the **SYN**?
 - What is the value of the Acknowledgement field in the **SYNACK** segment?
 - What is it in the segment that identifies the segment as a **SYNACK** segment?



Wireshark – Exercise 1



- Open **Wireshark**, then use the **File** menu and the **Open** command to open the file “**lab2.pcap**”.
- Look at the packets to familiarize with the Wireshark GUI and answer the following questions:
 - 1.What is the IP address of the client that initiates the conversation?
 - 2.Use the first two packets to identify the server that is going to be contacted. List the common name, and three IP addresses that can be used to reach the server.
 - 3.What are the port numbers involved in the Three-Way TCP Handshake (filter wireshark to observe TCP packets only)?

Do you have any questions?



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